

AUBURN-LEWISTON, ME, USA

WMO: 726184

Lat: 44.050N

Lon: 70.283W

Elev: 88

StdP: 100.27

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94709

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-17.7	-29.1	0.3	-17.1	-26.1	0.4	-15.6	11.0	-7.0	9.7	-6.7	1.7	10	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	31.0	21.9	28.8	21.1	27.3	19.8	23.5	28.5	22.3	26.9	21.3	25.2	4.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	16.9	26.5	21.0	15.8	25.1	19.8	14.7	23.8	70.0	28.4	65.9	26.6	62.0	25.0	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.2	7.3	-25.8	33.5	3.8	1.4	-28.5	34.5	-30.7	35.3	-32.9	36.1	-35.6	37.0
			-25.8	25.1	3.4	1.1	-28.2	25.9	-30.2	26.5	-32.1	27.1	-34.5	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-6.7	-5.0	-0.2	6.2	12.0	17.3	20.4	19.5	15.5	8.8	2.8	-3.0
	DBStd	10.34	6.10	5.73	5.27	4.05	3.86	3.79	2.89	2.84	3.92	4.16	4.68	5.35
	HDD10.0	2103	517	421	319	128	20	1	0	0	3	73	221	402
	HDD18.3	4188	775	653	576	364	199	64	12	19	103	295	467	660
	CDD10.0	1142	0	0	2	14	83	219	322	294	167	37	4	0
	CDD18.3	187	0	0	0	0	4	33	76	56	17	1	0	0
	CDH23.3	1761	0	0	4	12	88	327	669	490	161	10	0	0
	CDH26.7	473	0	0	1	2	25	90	183	128	43	1	0	0
Wind	WSAvg	2.7	2.8	2.9	3.3	3.2	2.8	2.4	2.2	2.1	2.2	2.6	2.6	2.7
Precipitation	PrecAvg	1135	86	83	101	103	93	90	84	76	86	103	118	108
	PrecMax	1675	263	197	304	222	290	282	185	275	336	332	295	320
	PrecMin	752	8	3	17	12	16	17	14	6	14	19	30	30
	PrecStd	202	49	37	52	44	56	51	38	45	51	60	54	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.0	19.1	24.9	30.1	32.2	32.6	32.5	31.2	24.8	17.6	12.7
		MCWB	9.6	6.9	11.9	15.5	18.9	22.5	23.7	21.7	22.0	17.4	12.5	10.0
	2%	DB	7.6	7.6	12.6	19.1	26.0	28.9	30.7	29.8	27.4	21.2	14.9	9.1
		MCWB	5.5	4.2	7.1	11.7	16.5	21.2	22.5	21.3	20.0	15.4	11.6	7.1
	5%	DB	4.1	5.1	9.0	16.3	22.5	27.2	28.7	27.7	24.9	18.0	12.7	7.2
		MCWB	1.9	2.2	5.0	9.4	14.8	19.6	21.5	20.2	18.8	13.9	10.0	4.7
	10%	DB	2.2	2.8	7.2	13.7	19.8	24.9	27.2	26.2	22.6	16.3	10.3	4.0
		MCWB	0.1	0.5	3.5	7.7	13.4	17.8	20.3	19.2	17.5	12.4	7.6	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	7.9	12.7	16.3	20.0	24.1	25.0	24.5	23.1	19.1	15.3	11.1
		MCDB	11.1	9.4	18.5	21.7	26.7	29.7	30.6	29.5	28.7	21.4	17.0	12.4
	2%	WB	6.1	5.0	8.3	13.5	17.8	22.4	23.6	22.9	21.4	16.8	12.7	7.7
		MCDB	7.6	7.3	10.9	18.3	23.0	27.6	28.6	27.8	25.0	19.3	14.0	8.7
	5%	WB	2.6	2.8	5.5	10.7	15.9	21.0	22.5	21.6	20.0	15.0	10.3	4.6
		MCDB	4.3	4.6	8.6	14.3	21.0	25.1	27.0	25.7	23.1	18.1	12.4	6.7
	10%	WB	0.2	0.9	3.6	8.6	14.1	19.4	21.4	20.5	18.6	13.0	7.9	2.4
		MCDB	1.7	2.9	6.4	13.0	18.6	23.2	25.4	24.3	21.4	15.7	9.6	4.0
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.3	9.9	11.0	11.4	11.3	11.3	11.9	11.9	10.9	9.2	9.2
		MCDBR	11.0	11.7	13.3	16.1	16.7	14.6	13.7	14.0	14.6	14.4	12.1	11.6
		MCWBR	9.8	9.1	8.9	9.5	8.9	7.9	6.6	7.1	8.2	9.5	9.5	9.8
	5% WB	MCDBR	11.0	10.7	12.2	14.1	14.1	12.7	11.5	11.9	11.9	12.4	10.4	11.0
		MCWBR	10.4	9.0	9.0	9.5	8.2	7.6	6.4	6.7	7.6	9.2	9.3	10.0
Clear-Sky Solar Irradiance	taub	0.276	0.293	0.326	0.351	0.380	0.432	0.445	0.412	0.372	0.332	0.315	0.285	
	taud	2.423	2.365	2.399	2.410	2.375	2.239	2.214	2.300	2.397	2.508	2.517	2.488	
	Ebn at Noon	859	902	913	913	891	842	825	841	852	849	807	815	
	Edh at Noon	73	92	103	111	118	136	138	123	103	81	68	63	
All-Sky Solar Radiation	RadAvg	1.63	2.51	3.67	4.55	5.09	5.49	5.70	5.19	4.21	2.66	1.72	1.28	
	RadStd	0.13	0.21	0.22	0.49	0.63	0.46	0.43	0.29	0.25	0.24	0.16	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+72	N/A

Nomenclature: See separate page